

Literature Review: SentinelAI

Author(s)	Title / Focus	Methodology	Key Findings	Relevance to Proposed System
Carlini, N., et al. (2021)	<i>Extracting Training Data from Large Language Models / LLM privacy risks</i>	Conducted extraction attacks on GPT-2 to demonstrate that LLMs memorize and regurgitate PII and credentials found in their training data.	Proven that LLMs expose sensitive data inadvertently, highlighting the critical necessity to sanitize inputs/outputs.	Justifies the core need for SentinelAI to intercept and block PII from being sent to LLMs, preventing accidental data exposure.
Greshake, K., et al. (2023)	<i>A Comprehensive Analysis of Novel Prompt Injection Threats / Security of AI application workflows</i>	Investigated how malicious or highly sensitive prompts can leak data and compromise LLM-integrated applications.	LLMs are highly susceptible to indirect prompt vulnerabilities that can silently leak user tokens and PII to third-party endpoints.	Highlights the necessity of a pre-submission scanning pipeline to neutralize risky string patterns before they reach the AI model.
Yao, Y., et al. (2023)	<i>Cloud Security Posture and Prompt Privacy / Cloud API and LLM telemetry</i>	Evaluated how third-party AI APIs (like OpenAI and Anthropic) handle enterprise telemetry and prompt logging.	By default, unstructured prompts sent to third-party endpoints are logged and can be reviewed, risking severe enterprise compliance breaches.	Defines the exact problem SentinelAI solves by stopping PAN, Aadhaar, and API Keys at the browser-level <i>before</i> the network request fires.
Kang, D., et al. (2023)	<i>Exploiting Programmatic Behavior of LLM Agents / Hardcoded credentials in prompts</i>	Tested how AI coding assistants and Chatbots handle injected source code and API keys blindly pasted by developers.	Developers frequently paste massive blocks of raw code containing hardcoded credentials (AWS keys, DB strings) into models like ChatGPT or Copilot.	Validates the urgent need for SentinelAI's source code classifier module and its specific regex rules targeting AWS Keys, JWTs, and DB URIs.
Shabtai, A., et al. (2012)	<i>Securing IT Environments against Data Leakage / Traditional DLP techniques</i>	Surveyed host-based and network-based DLP algorithms, heavily comparing regular expressions versus complex context fingerprinting.	Regular expressions and heuristic scanning remain the most computationally efficient methodologies for real-time endpoint interception.	Scientifically validates SentinelAI's use of optimized, client-side regex patterns running natively inside a browser extension for zero-latency scanning.
Lison, P., et al. (2021)	<i>Named Entity Recognition for Privacy Protection / Machine Learning PII detection</i>	Applied NLP and trained BERT models to identify specific secure entities like names, SSNs, and banking details in unstructured text.	ML models are highly accurate but suffer from high processing latency and compute overhead compared to deterministic rule engines.	Informed SentinelAI's design choice to rely primarily on blazing-fast regex in the browser, while reserving ML tracking for the backend pipeline.
Bandhakavi, S., et al. (2010)	<i>Vetting Browser Extensions for Security Vulnerabilities / Extension architecture</i>	Analyzed Chrome's extension framework and how permissions and interceptors ought to interact securely via the DOM and background scripts.	Extensions must strictly isolate DOM interaction (content scripts) from heavy data logic (background scripts) to maintain security.	Perfectly aligns with SentinelAI's architecture where the <code>content.js</code> script only scrapes the DOM while <code>background.js</code> securely manages API routing.
Mireshghallah, F., et al. (2022)	<i>Privacy and Security in Deep Learning / Empirical text sanitization techniques</i>	Tested various text sanitization techniques (masking vs. redaction vs. outright blocking) prior to executing AI model inference.	Total blocking of sensitive payloads is far more foolproof than "masking" entities, as LLMs can often logically infer masked data from context.	Reinforces SentinelAI's aggressive BLOCK mechanism (clearing the <code>textarea</code> entirely) instead of just attempting to replace sensitive words with asterisks.
Elazari, A. (2018)	<i>Regulatory Compliance and Data Loss Prevention / Legal DLP frameworks</i>	Analyzed GDPR, HIPAA, and modern DPDP data protection frameworks and their technological enterprise requirements.	Enterprises must demonstrate proactive "data in motion" filtering and maintain strict audit logs to avoid massive compliance fines.	Dictates the necessity of SentinelAI's Vercel-based Audit Dashboard, allowing enterprise admins to satisfy compliance by tracking interception metadata.
Ouyang, L., et al. (2022)	<i>Training language models to follow instructions with human feedback / RLHF capabilities</i>	Documented how modern chatbots process instructions, often absorbing any provided user context directly into their active context window for continuous reference.	Instruct-models implicitly trust the context provided by users, meaning any leaked PII becomes part of that active AI "conversation memory."	Emphasizes why SentinelAI intercepts prompts at the exact moment the <code>Enter</code> key is pressed—preventing sensitive data from ever becoming conversational memory.